A Review: Weighted Quantile Sum (WQS) regression and its extensions

Sili Fan

CTSC UC Davis Health

slfan@ucdavis.edu

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Overview

1. Background & Motivations
2. Method: WQS regression
3. Extensions
4. Summary

Background & Motivation

- Exposomics is a field of study that focuses on the analysis of environmental exposures and their effects on human health over a person's lifespan.
- Traditionally researchers examine the impact of a single exposure (e.g., O_3, NO_x, SO_2 ,Pb, CO, PM).
- A mixture of exposures is of current interest (additive or synergic effect).
- Exposures are highly correlated. Exposomics data can be high-dimensional.

Background & Motivation

- Limitation of traditional statistical methods:
  - General linear regressions suffer from multi-collinearity.
  - General linear regressions suffer from $p >> n$.
  - LASSO tends to select arbitrary members from the group of (correlated) predictors, leading to poor specificity in variable selection.
  - Elastic net encourages a "grouping effect", either including or eliminating the whole group, which is not desired in exposomics.

Method: WQS regression

Carrico et al. (2015) proposed the following weighted quantile sum regression model

$$g(\mu) = \beta_0 + \beta_1 \sum_{j=1}^{c} w_j q_j + \mathbf{z}'\boldsymbol{\phi},$$

where c is the number of exposures, w_j's are weighted parameters (subjecting to $0 \leq w_j \leq 1$, $\sum_{j=1}^{c} w_j = 1$), q_j's are quantile scores, and $\mathbf{z}$ is the vector of confounders. $g(\cdot)$ is a link function and $\mu = \mathbf{E}(y)$. y is from an exponential family. Let $\theta = (\beta_0, \beta_1, w_1, \ldots, w_c, \boldsymbol{\phi})'$ be the vector of parameters to be estimated.

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WQS regression can deal with multi-collinearity, high-dimensionality, and outlier effect.

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- Basic idea is to split the dataset into a training set N_t and a validation set N_v.
  1. N_t is used to estimate the w_j's
  2. N_v is used to estimate the β_0, β_1 and $\boldsymbol{\phi}$.

Method: WQS regression

- Estimation (w_j's):
  - To increase the sensitivity in detecting the important predictors, B bootstrap samples are generated from N_t, inspired by Stability Selection - Meinshausen et, al (2010).
  - For each bootstrap samples, get maximum likelihood estimator of $\theta^{(b)}$

$$\hat{\theta}^{(b)} := \arg\max_{\theta} \left[l(\theta; y^{(b)}) + \lambda \left(\sum_{j=1}^{c} w_j - 1 \right) \right]$$

  - Let $\hat{w}_j = \frac{1}{B} \sum_{b=1}^{B} \hat{w}_j^{(b)} f(\hat{\beta}_1^{(b)})$, where $f(\cdot)$ is a signal function, giving higher value to the bth bootstrap sample if $\hat{\beta}_1^{(b)}$ has higher signal.

Method: WQS regression

- Estimation (β):
  - Define WQS index as WQS $:= \sum_{j=1}^{c} \hat{w}_j q_j$
  - Use N_v to estimate β_0, β_1 and $\boldsymbol{\phi}$, via model

$$g(\mu) = \beta_0 + \beta_1 \text{WQS} + \mathbf{z}'\boldsymbol{\phi}$$

- A test for significance of β_1 (using N_v) determines whether there is an association between the group of exposures and the outcome variable.
- The significance of β_1 permits the further interpretation of the w_j's, deemed as the importance of the jth exposure.

Method: WQS regression

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The advantages of the WQS regression are

- Extremely general and has a very wide range of applicability.
- Deal with multi-collinearity (to some extend).
- Robust with outliers.
- Four simulation studies in the paper showed WQS has a consistent higher sensitivity and higher specificity in two of them, compared with ordinary and logistic regressions and LASSO-type methods.

Extensions

- Random Selection WQS [Curtin et al. (2021)]
- Repeated holdout validation WQS [Tanner, Bornehag, and Gennings (2019)]
- Bayesian WQS [Colicino et al. (2020)]
- Lagged WQS [Gennings et al. (2020)]

Extensions: Random Selection WQS

- WQS still suffer in $p >> n$ scenario
- A novel implementation of WQS regression, WQS_{RS} for high-dimensional mixtures with highly correlated components was proposed in Curtin et al. (2021).
- This approach added a procedure of random selection on exposures for each bootstrap sample for parameter estimation.
- Basically, in each bootstrap sample, a s-sized subset of exposures are selected, where $s < n < c$. And

$$\hat{\boldsymbol{w}}_{(c\times 1)} = \frac{1}{B}\sum_{b=1}^{B}\left(A^{(b)}\right)'_{(C\times s)}\hat{\boldsymbol{w}}^{(b)}_{(s\times 1)},$$

where $A^{(b)}$'s are indicator matrices.

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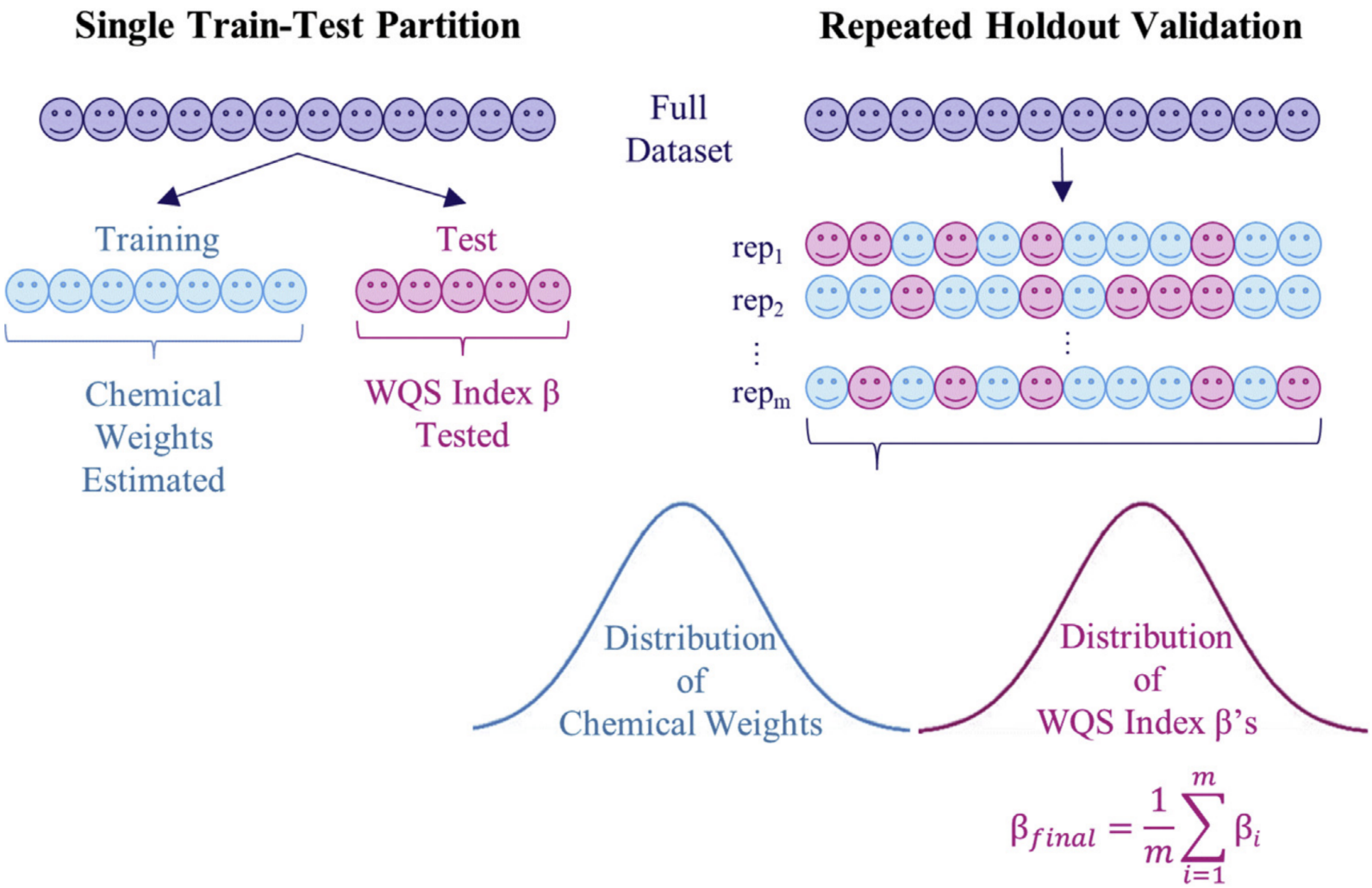
  where $A^{(b)}$'s are indicator matrices.

- It is shown to be more effective compared to WQS in modeling contexts with large predictor sets, complex correlation structures, or where the numbers of predictors exceeds the number of subjects.

Extensions: Repeated holdout validated WQS

- The partitioning of dataset into training and validation subset in the WQS estimation reduces statistical power.
- This partition can also lead to unrepresentative sets of data and unstable parameter estimates.
- Tanner, Bornehag, and Gennings (2019) showed that conducting a WQS on the full dataset without partitioning produces optimistic results and proposed to apply a repeated holdout validation combining cross-validation and bootstrap resampling. This procedure can be summarized in the following picture:

Extensions: Repeated holdout validated WQS



Extensions: Repeated holdout validated WQS

- The inference on β_1 is based on the repeated holdout sets.
- Also, it allows the investigator to characterize weight uncertainty, as shown in the following picture,

Extensions: Repeated holdout validated WQS

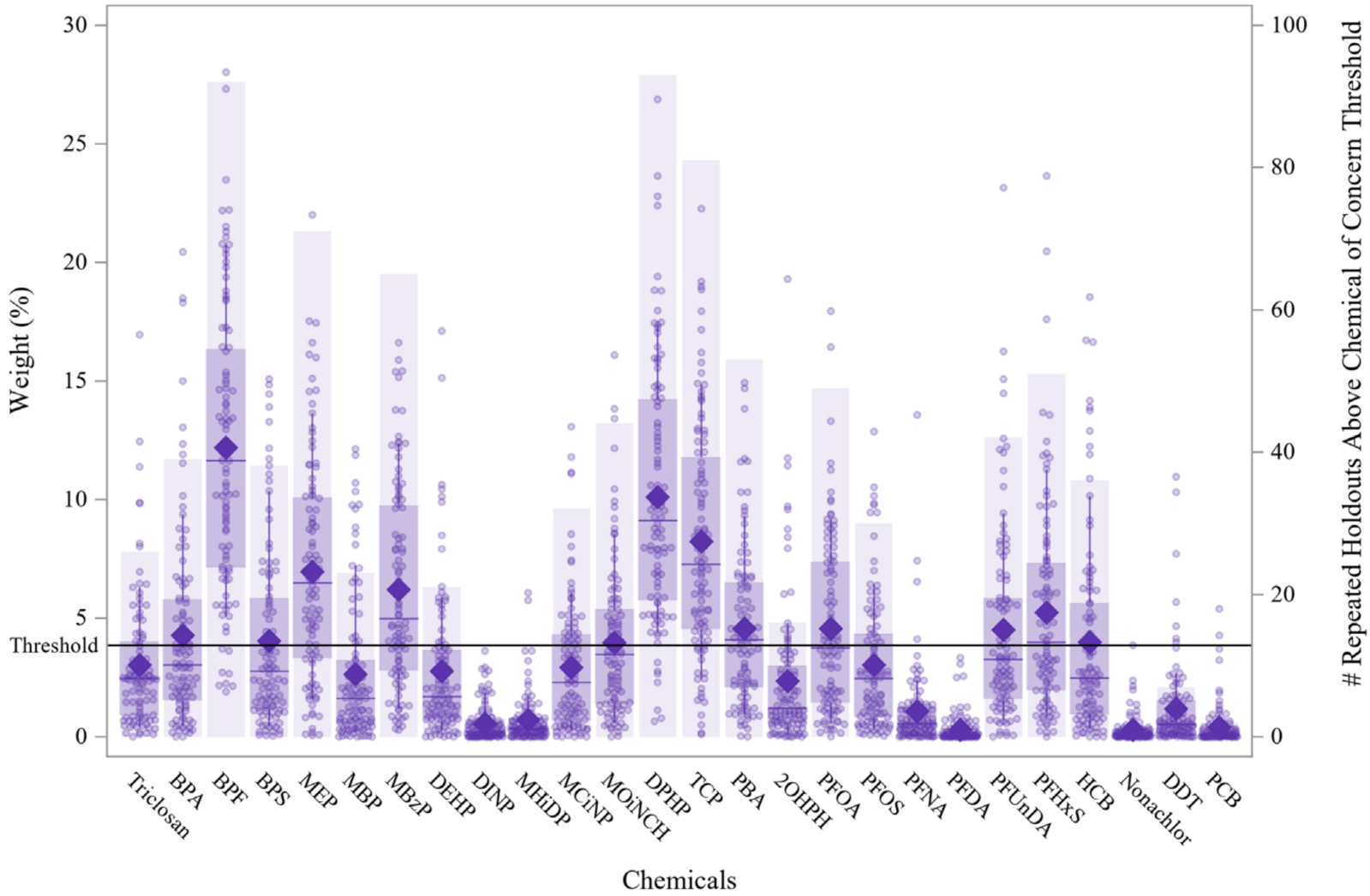


Fig. 2. Chemicals of Concern Identification & Uncertainty for 26 Endocrine Disrupting Chemicals in Relation to IQ.

Extensions: Repeated holdout validated WQS

- Through this procedure they obtained distributions of parameters to make inference on parameters.
- It increased the reproducibility of the analysis.
- A limit of this approach is the higher computational intensity.

Extensions: Bayesian WQS

- To extend WQS to a Bayesian framework, Colicino et al. (2020) proposed Bayesian WQS.
- Basic idea:
  1. Put prior to all the parameters, i.e. normal distribution for β_0, β_1 and ϕ, Dirichlet distribution for $\boldsymbol{w}$;
  2. Use numeric methods, e.g. Hamiltonian-Monte Carlo to obtain the final distribution (for β_1).
- The specifics aren't displayed, but this is included to illustrate how WQS is protentially capable of incorporating the Bayesian world.

Extensions: Lagged WQS

- In exposomics, lagging effects refer to the time delay between the exposure to a certain environmental factor and the onset of the health effect.
- "The developmental timing of exposures to toxic chemicals or combinations of chemicals may be as important as the dosage itself. This concept is called *critical windows of exposure*."
- Gennings et al. (2020) proposed the lagged WQS regression to find the critical window.

```
#>    ID sex time        out      pred1      pred2      pred3      pred4      pred5
#> 1   1   1    1 -0.8322414 -0.2062299 1.30295010 -1.3870911  0.1607504 -0.1757468
#> 12  1   1    2 -0.8322414 -0.0941091 0.42957397 -0.4552804  0.3642088 -1.1884153
#> 23  1   1    3 -0.8322414  2.9357351 0.01995526  0.7725593  0.8464955 -1.4320769
#> 25  1   1    4 -0.8322414 -1.3437874 0.86728598  0.7679771  0.1602173 -0.7590517
#> 26  1   1    5 -0.8322414  0.7142656 0.30884399  0.7302721  0.1933808  0.3498828
#> 27  1   1    6 -0.8322414  0.6871639 1.26009045  0.9717983 -0.0763416  0.6857857
```

Extensions: Lagged WQS

- Procedure:
  1. Determine a temporal domain and time unit (e.g., a week) for analysis.
  2. For each time unit, conduct $\text{WQS}(t) := \sum_{j=1}^{c} w_j(t) q_j(t)$.
  3. For each time unit, estimate the time varying association across these indices using genrealized additive mixed effect model (GAMM):

$$\text{WQS}(t) = \beta_0(t) + \beta_1(t) Y + \boldsymbol{z}'\boldsymbol{\phi} + u + \epsilon,$$

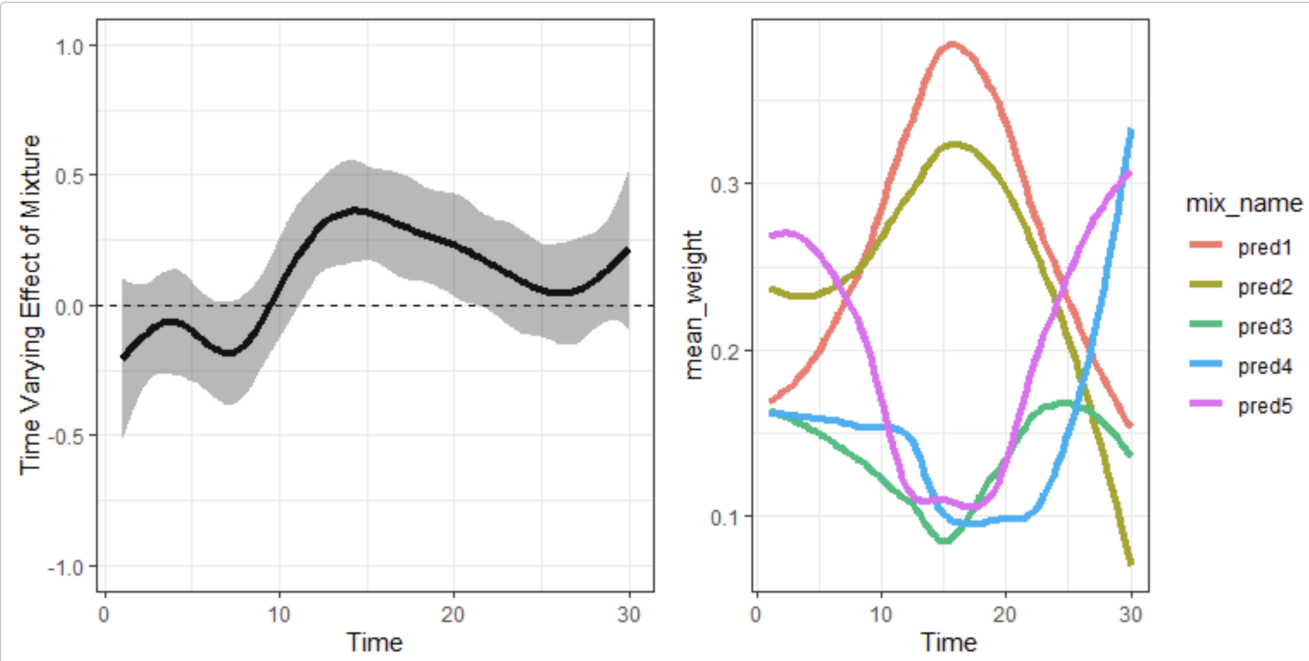
     where u is the random intercept term. Note the switch of $\text{WQS}(t)$ and Y.
  4. Conduct a smooth function (with confidence band) on $\beta_1(t)$ across t. OR use bootstrap to get the confidence interval.

- Lagged WQS results in two plots.

Extensions: Lagged WQS

- Left Panel:
  The time-varying coefficient. Time periods whose confidence bands are not overlapping with zero are interpreted as *critical window*.

- Right Panel:
  In the *critical window*, the weights represent the contribution of the exposures.



Summary

- WQS is a useful model in exposomics to handle data with multi-collinearity, high-dimensionality and outlier effects.

- Four extensions.
  - Random Selection WQS deals with high-dimensionality.
  - Repeated holdout validation WQS improves the statistical power and reproducibility.
  - Bayesian WQS connects to the Bayesian world.
  - Lagged WQS examines the lagging effect of exposures on an outcome.

- Critiques (IMO):
  1. Does the optimization converge? Recall

$$g(\mu) = \beta_0 + \beta_1 \sum_{j=1}^{c} w_j q_j + \mathbf{z}'\boldsymbol{\phi}$$

  2. Statistics properties are not yet studied.

Thank you!

- Questions?